



Inspiring Excellence

Assignment Cover Page

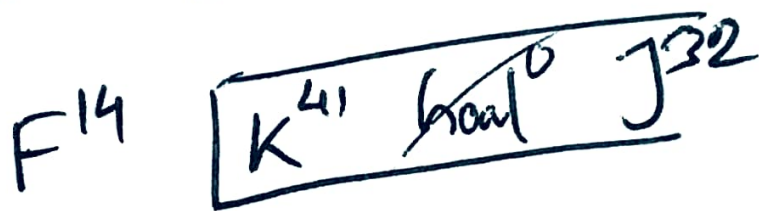
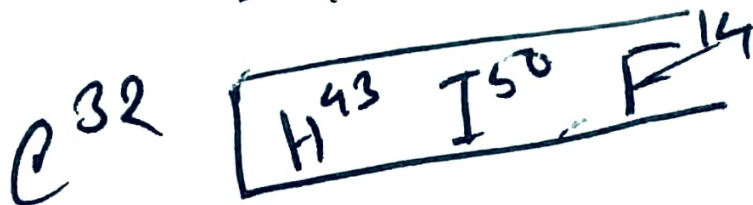
CSE422

Artificial Intelligence

Course Code	CSE422
Course Title	Artificial Intelligence
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Section	18

Assignment 1

1. a) BGFS



Goal⁰

cost (start $\rightarrow$ C $\rightarrow$ F $\rightarrow$ Goal)

$$= 0 + 20 + 99 + 50 = 169$$

b) A* search

A*

A* B⁹² C⁵² L¹⁰⁴

C⁵² B⁹² B²⁹⁵ H²⁸⁴ I¹⁸¹ F¹²³ L¹¹⁶

B⁹² C²⁹⁹ D⁹³ H²⁸⁴

D⁹³ H³⁰² E¹⁴⁹

L¹⁰⁴ C¹²⁰ F⁷⁴ K¹⁰⁹

F⁷⁴ C¹⁰¹ I²⁰⁹ J¹³² Goal¹¹⁰ K¹⁴³

K¹⁰⁹ F¹²⁴ Goal¹⁰⁹

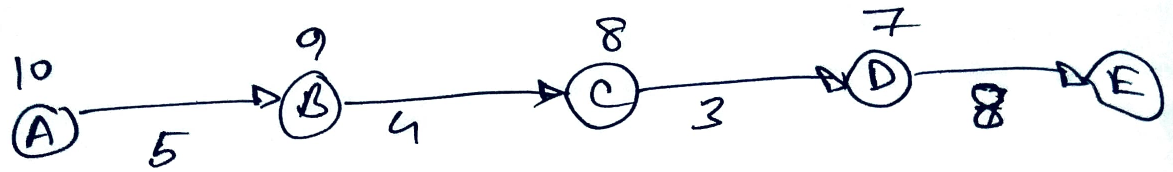
Goal¹⁰⁹

— X —

A → L → K → Goal

$$\text{Cost} = 0 + 48 + 20 + 41 = 109$$

2)



consistency $h(n) \leq h(n, n') + h(n')$

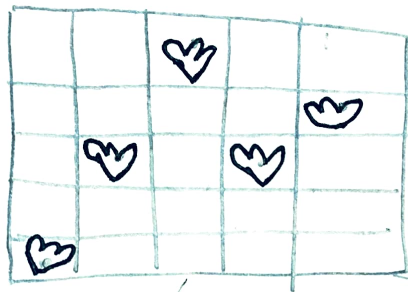
$$h(A) : 10 \leq 5 + 6$$

$$h(B) : 9 \leq 4 + 8$$

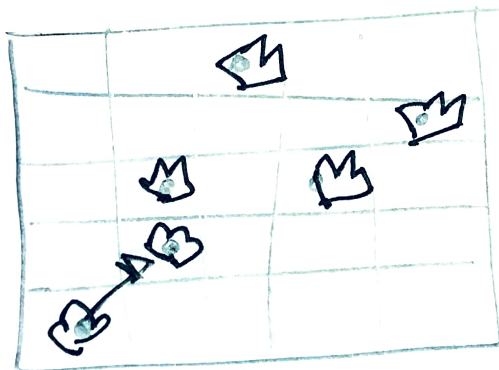
$$h(C) : 8 \leq 3 + 7$$

$$h(D) : 7 \leq 8 + 0$$

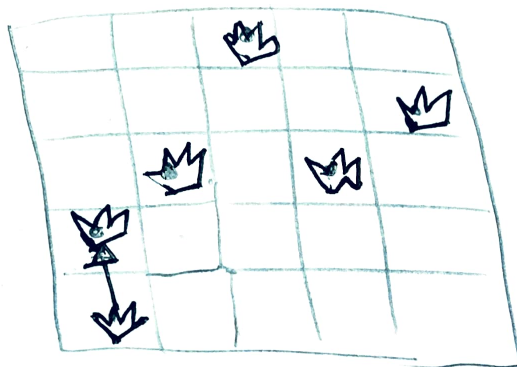
3)



attaching pair = 2



attaching pair = 3



attaching pair = 3

8 puzzle

1	2	3
4	—	6
5	8	7

$$\text{misplaced} = 1 + 2 = 3$$

1	—	3
4	2	6
5	8	7

$$\begin{aligned}\text{mis} &= 1 + 1 + 2 \\ &= 4\end{aligned}$$

1	2	3
—	4	6
5	8	7

$$\begin{aligned}\text{mis} &= 2 + 2 \\ &= 4\end{aligned}$$

1	2	3
4	6	—
5	8	7

$$\text{mis} = 2 + 2 = 4$$

1	2	3
4	8	6
5	—	7

$$\begin{aligned}\text{mis} &= 1 + 3 \\ &= 4\end{aligned}$$

So, this is local maxima because going to either to moves from 5 queen or 8 tiles, they would go to lower value away from goal, that is, going to lower steep.

Q) In simulated annealing, temperature runs the probabilistic optimization, meaning algorithm's behavior to explore or stop for a neighbor. hill-climb algorithm is determined.

Firstly, the ~~temp run~~ SA runs as long as it doesn't come from high to low temperature value.

Secondly, $e^{\Delta E/T}$ gives the probability of taking worse solutions when temperature high but low temperature gives rarely taking worse situations. So, temperature is the main probabilistic factor.